



POWER BI MAJOR CAPSTONE PROJECT
END TO END
POWER BI PROJECT GUIDE

PROJECT TITLE SUPERSTORE SALES & PROFIT ANALYTICS
DASHBOARD

PROJECT Sales Analytics
CATEGORY Profit Analytics
Business Intelligence

INSTITUTE NAME Infinity Code Nexus

TRAINER / GUIDE Mr. Anand Kumar

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TOOLS USED Power BI Desktop, Power Query, DAX & Power BI
Service

SUBMISSION DATE [21/06/2026]

Prepared as part of the Power BI Masterclass Program to demonstrate skills in
Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX,
Storytelling, and Business Intelligence Reporting.

PROJECT OVERVIEW

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PROJECT OBJECTIVE

Develop a comprehensive Power BI dashboard to analyze sales, profit, customers, products, and regional performance for a retail business

Participants are expected to:

- ✓ Analyze sales performance
- ✓ Evaluate profitability
- ✓ Understand customer behavior
- ✓ Assess regional performance
- ✓ Generate business insights
- ✓ Present recommendations

PROJECT GOAL

Transform transactional retail data into meaningful business insights that support decision-making, performance monitoring, and business growth.

BUSINESS PROBLEM

Retail management requires a clear understanding of sales performance, profitability, customer behavior, and product contribution to make better business decisions.

The objective is to identify:

- **Which products generate the highest sales?**

The highest sales were generated by **Accessories**, contributing approximately **36.40%** of total revenue. This indicates strong customer demand and highlights the product category as a key revenue driver for the business.

- **Which regions contribute the most profit?**

The **West** region contributed the highest profit, generating **\$108.42k** with a profit margin of **0.15%**. This suggests strong market performance and effective sales strategies in that region.

- **Which customer segments drive business growth?**

The **Consumer** segment contributed the largest share of sales and profit, accounting for **50.56%** of total revenue. This indicates that the segment is a

major driver of business growth and should remain a focus for marketing and customer retention efforts.

- **How discounts impact profit?**

The analysis shows that as discounts increase, profit margins tend to decrease. Products with higher discount rates often generate more sales volume but contribute lower profitability, indicating the need for an optimized discount strategy.

- **Which products require business attention?**

Products such as **Furniture** require business attention because they generate low profit or incur losses despite contributing to sales. Management should review pricing, discount policies, and product costs to improve performance.

BUSINESS QUESTIONS

Participants must answer:

1. **What is total sales revenue?**

The total sales revenue generated during the analysis period was **\$2297200**. This represents the overall revenue earned from all customer orders and serves as a key indicator of business performance.

2. **What is total profit?**

The business achieved a total profit of **\$286397**, indicating the net earnings after accounting for costs and operational expenses. This metric helps evaluate overall profitability and financial health.

3. **Which products perform best?**

Accessories emerged as the best-performing products, generating the highest sales and/or profit. Its strong performance highlights high customer demand and significant contribution to business revenue.

4. **Which regions generate maximum revenue?**

The **West** region generated the highest revenue, contributing **\$725457** and accounting for **31.58%** of total sales. This indicates strong market presence and customer demand in that region.

5. **Which customer segments contribute most?**

The **Consumer** segment contributed the highest share of sales and profit, making it the primary driver of business growth. This segment represents the most valuable customer base for the organization.

6. **How do discounts affect profit?**

The analysis shows that higher discounts generally reduce profit margins. While discounts may increase sales volume, excessive discounting can negatively impact profitability, highlighting the need for a balanced pricing strategy.

7. **What are the sales and profit trends over time?**

Sales and Profit exhibited an overall positive growth trend over the analysis period, although some fluctuations were observed. The business experienced

peak performance during **November ,2017**, providing insights for future planning and resource allocation.

DATASET INFORMATION

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DATASET NAME : Superstore Dataset

DATASET DETAILS : Approximate Records 9,994+

Columns : 19+

EXPECTED ANALYSIS AREAS

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Sales Analytics

Profit Analytics

Customer Analytics

Product Analytics

Regional Analytics

Shipping Analytics

PROJECT OBJECTIVES

OBJECTIVE-1

Sales Performance Analysis : Analyze revenue generation and sales performance across regions.

OBJECTIVE-2

Profitability Analysis : Evaluate profit contribution and identify profit-driving areas.

OBJECTIVE-3

Customer Analysis : Understand customer segments and purchasing behavior.

OBJECTIVE-4

Product Analysis : Identify top-performing and low-performing products.

OBJECTIVE-5

Regional Analysis : Compare regional performance and business contribution.

OBJECTIVE-6

Executive Reporting : Create dashboards suitable for business leaders, managers, and decision makers.

EXPECTED OUTCOME

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Participants should demonstrate:

- ✓ Data Understanding
- ✓ Data Analysis
- ✓ KPI Development
- ✓ Dashboard Design
- ✓ Storytelling
- ✓ Business Recommendations

DASHBOARD REQUIREMENTS

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Participants must create a minimum of 5 Dashboard Pages covering all major business areas.

1.EXECUTIVE OVERVIEW

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Purpose: Provide a high-level summary of sales performance, profitability, and overall business health.

Required KPIs

- ✓ Total Sales
- ✓ Total Profit
- ✓ Profit Margin %
- ✓ Total Orders
- ✓ Average Order Value
- ✓ Top Performing Region

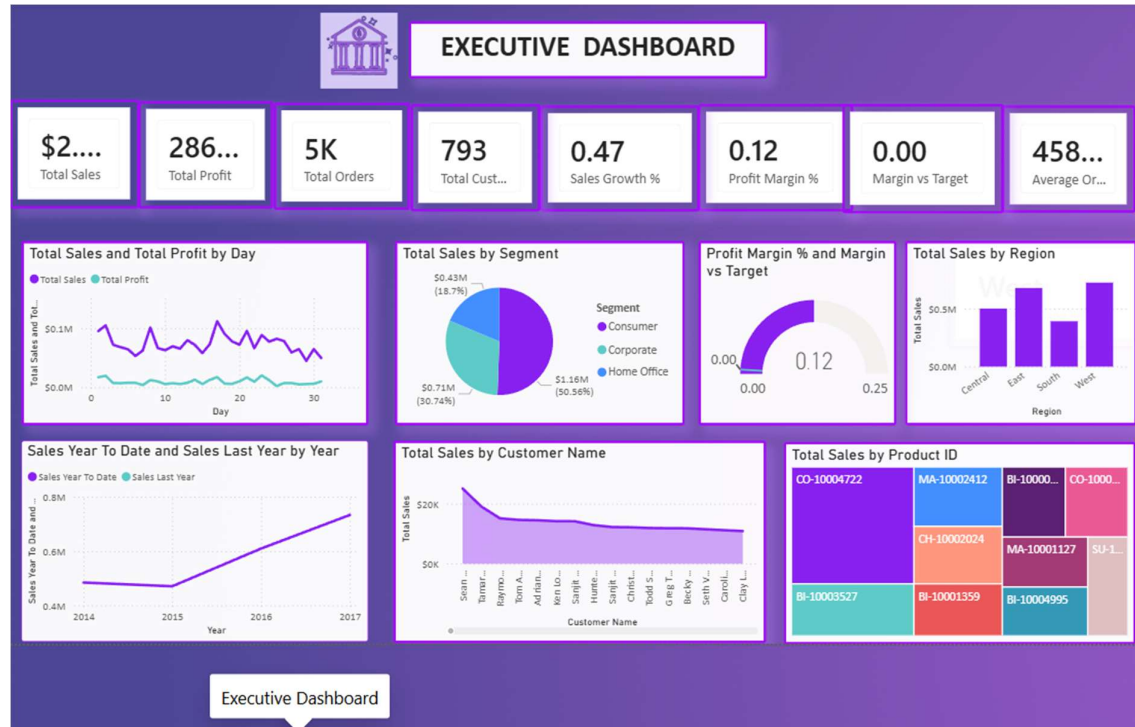
Suggested Visuals

Monthly Sales Trend

Sales by Region

Sales by Segment

Top Products

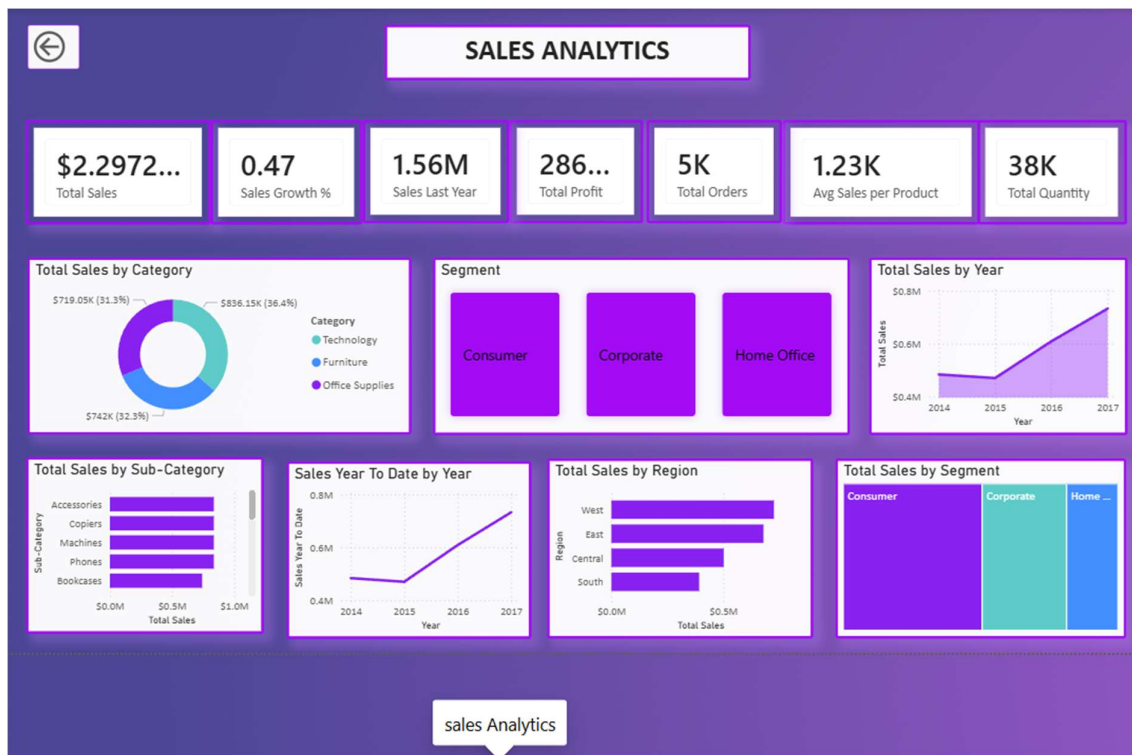


2.SALES ANALYTICS

Purpose: Analyze sales performance across products, regions, and customer segments.

Required Analysis :

- ✓Sales by Region
- ✓Sales by Category
- ✓Sales by Sub-Category
- ✓Sales Trend Over Time
- ✓Sales by Customer Segment

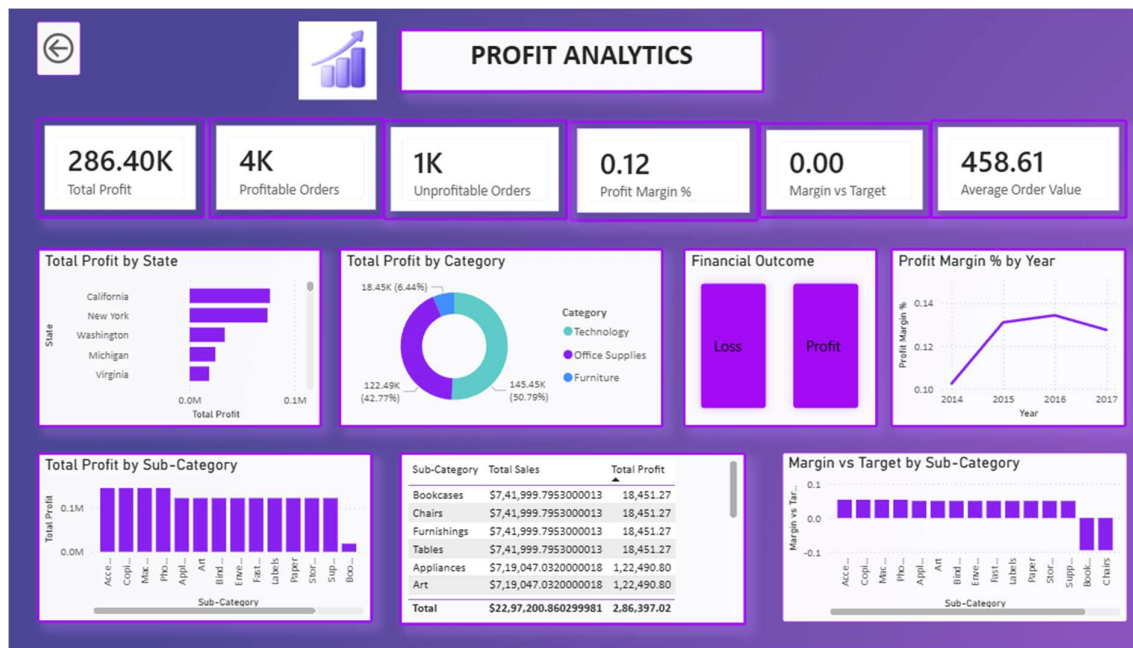


3.PROFIT ANALYTICS

Purpose: Understand profitability and identify profit drivers.

Required Analysis

- ✓Profit by Region
- ✓Profit by Category
- ✓Profit by Sub-Category
- ✓Profit Margin %
- ✓Top & Bottom Profit Products

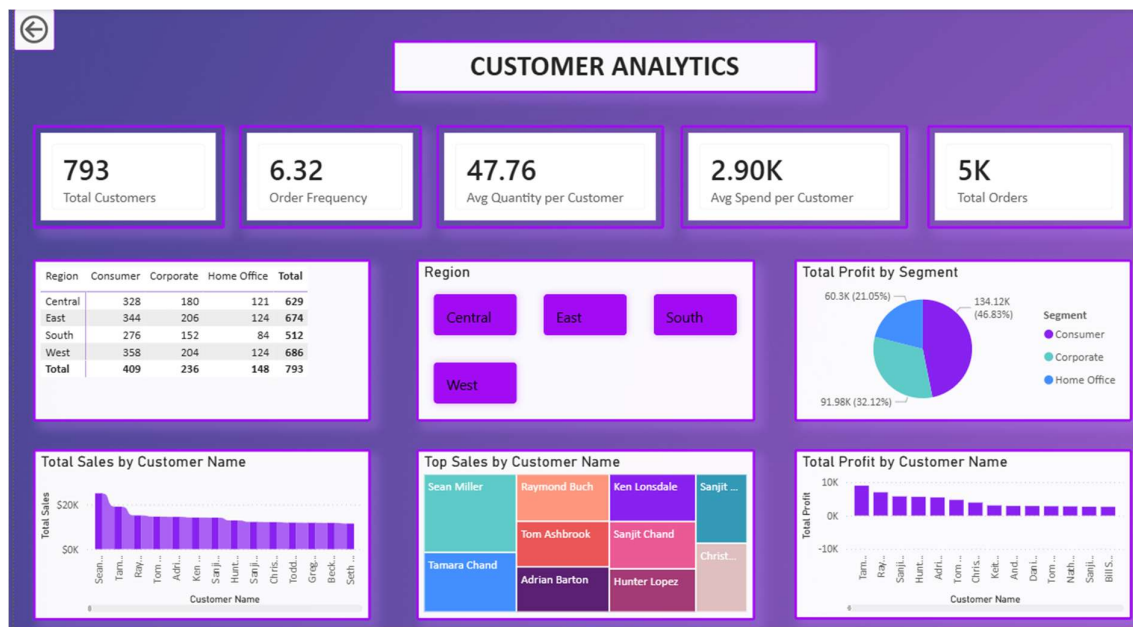


4.CUSTOMER ANALYTICS

Purpose: Analyze customer segments and purchasing behavior.

Required Analysis

- ✓Top Customers
- ✓Sales by Customer Segment
- ✓Customer Contribution
- ✓Regional Customer Analysis
- ✓Customer Purchase Patterns

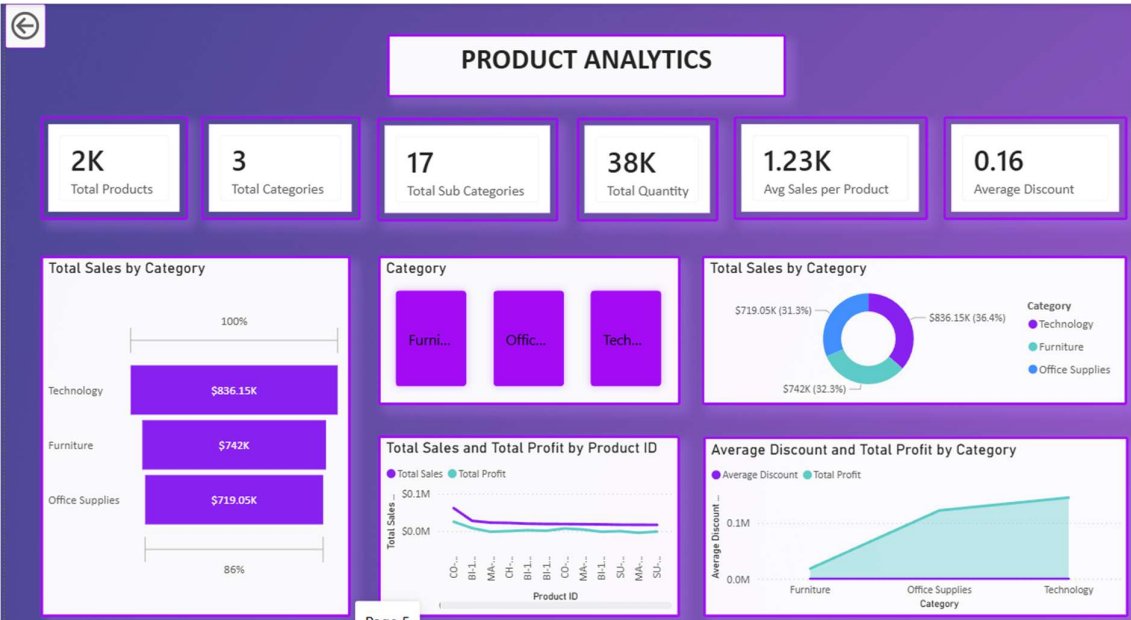


5.PRODUCT ANALYTICS

Purpose: Evaluate product performance and business contribution.

Required Analysis

- ✓Top Products by Sales
- ✓Top Products by Profit
- ✓Category Contribution
- ✓Sub-Category Analysis
- ✓Discount Impact on Profit



MANDATORY FEATURES

Participants must implement the following features.

DRILL DOWN

Examples:

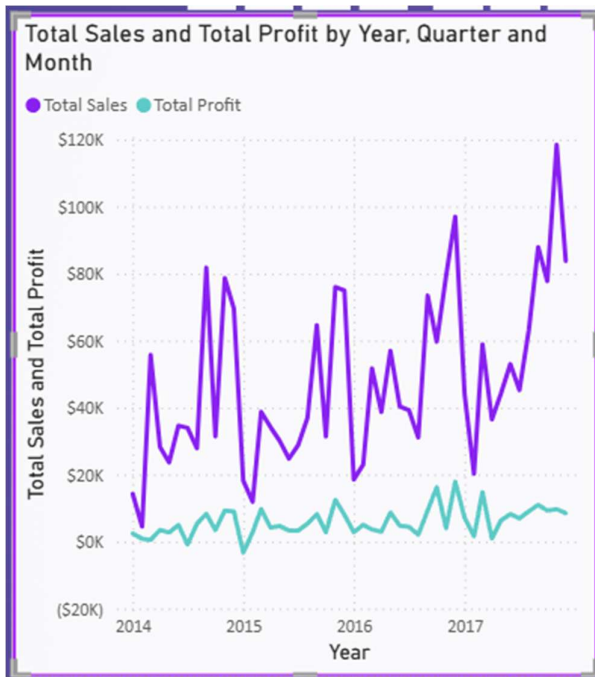
Year

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Quarter

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Month



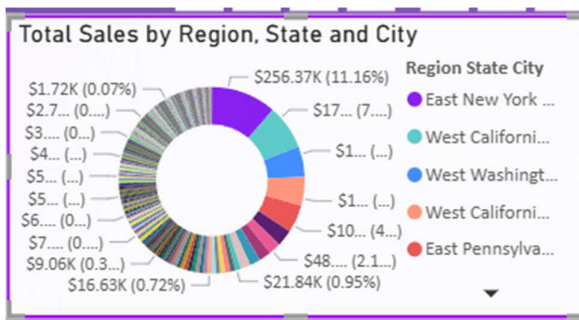
Region

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State

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City



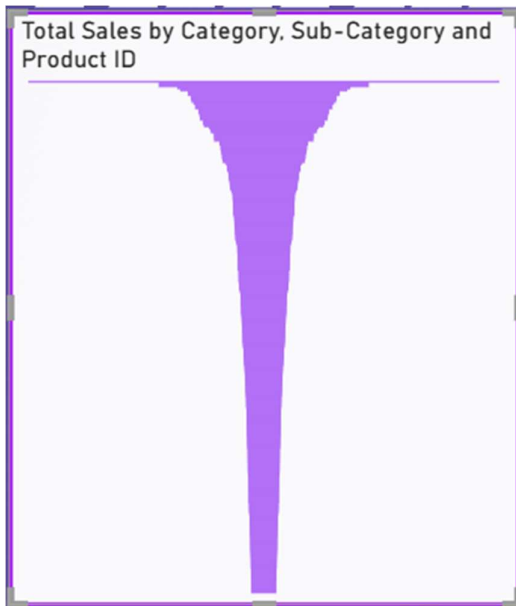
Category

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Sub-Category

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Product



DRILL THROUGH

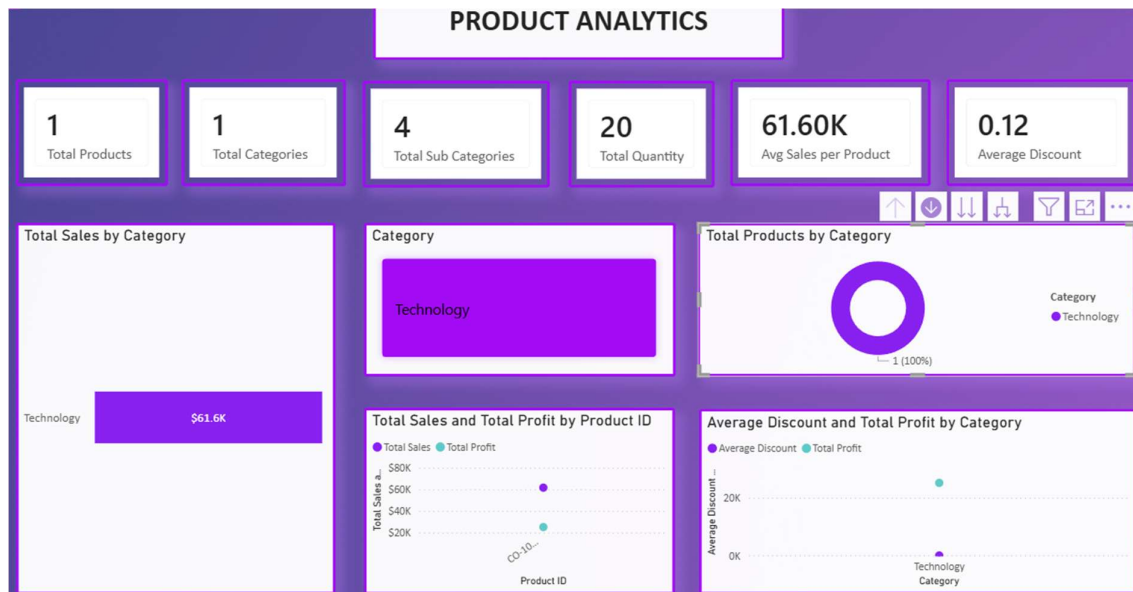
Examples:

Product



Product Detail Page

TITLE : PRODUCT ID : CO-10004722 detailed Analysis

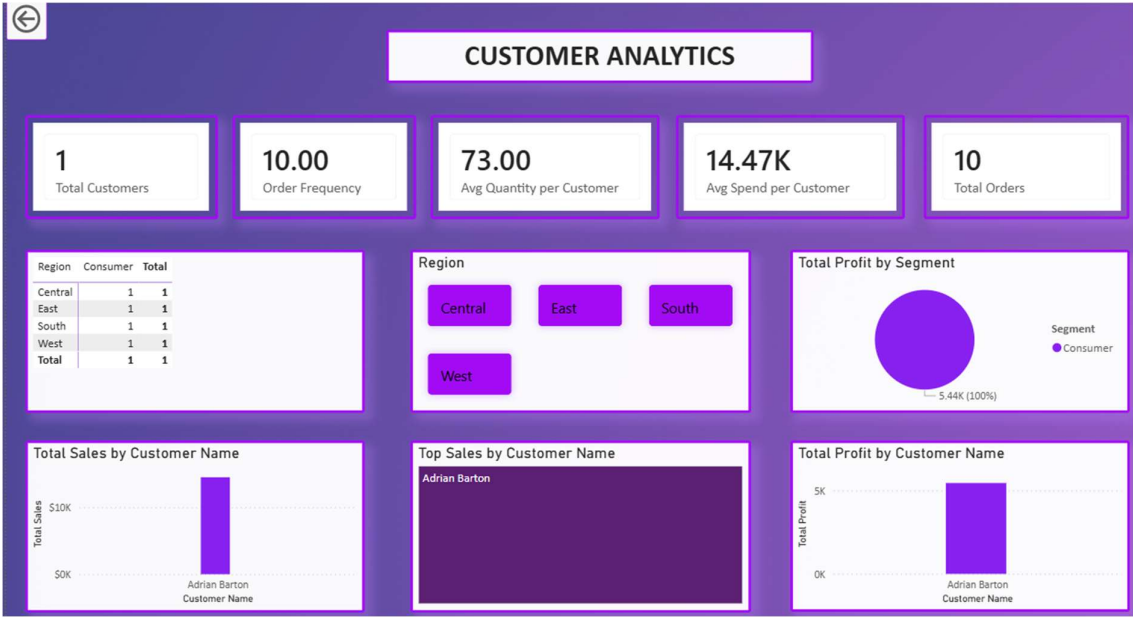


Customer



Customer Profile Page

TITLE : Customer Name: Adrian Baston detailed Analysis

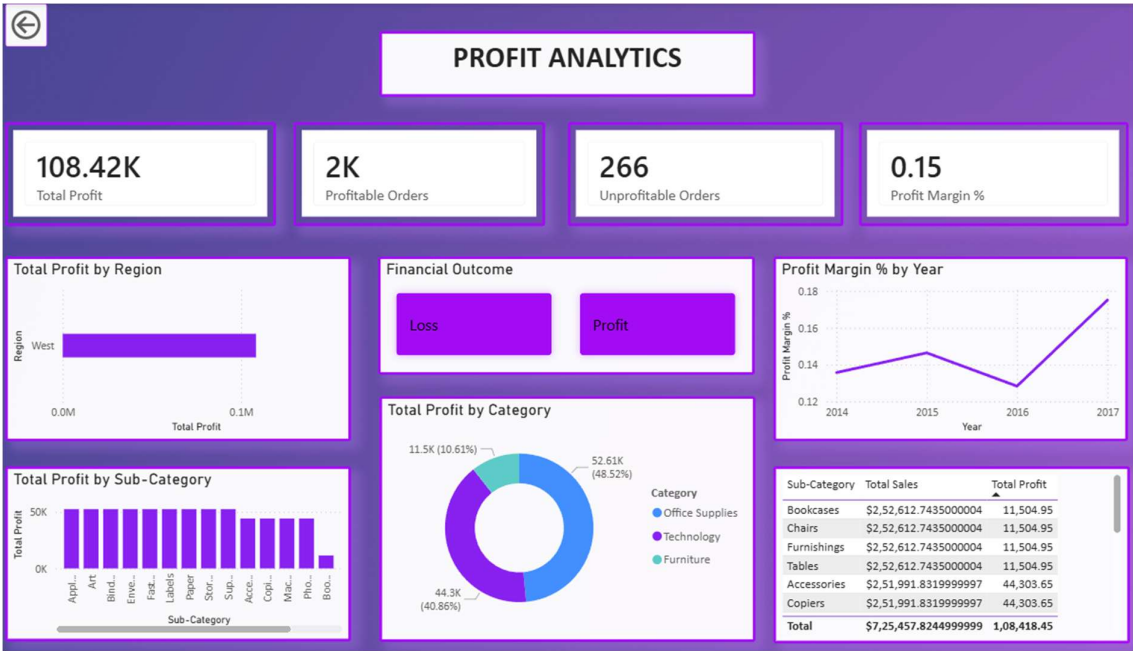


Region



Regional Analysis Page

TITLE : West Region detailed Analysis



TOOLTIPS

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Create custom tooltips for important visuals.

CONDITIONAL FORMATTING

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Apply color logic using DAX.

Green : Good Performance

Orange : Attention Required

Red : Negative Performance

DYNAMIC TITLES

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Titles must respond to slicer selections.

SLICERS

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Required Slicers

✓Year

✓Region

✓Category

✓Segment

PROJECT DELIVERABLES

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Participants must submit a complete Power BI project package consisting of the final PBIX file with all dashboard pages, interactions, drill-through, drill-down, and reporting features.

Project documentation must include the project objective, business problem, dataset overview, KPIs created, insights generated, and business recommendations.

Screenshots of all dashboard pages must be included covering Executive Overview, Sales Analytics, Profit Analytics, Customer Analytics, and Product Analytics dashboards.

A GitHub repository must be created containing the PBIX file, project documentation, screenshots, and a properly structured README file.

Participants must also deliver a professional presentation of 5–10 minutes explaining the analysis, insights, and recommendations.

JUDGING CRITERIA

- ✓Accuracy
- ✓Creativity
- ✓Business Understanding
- ✓Professional Design
- ✓Insights Generated

SUBMISSION GUIDELINES

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Participants must submit all project deliverables in an organized and professional manner. The final submission should include the PBIX file, project documentation, dashboard screenshots, GitHub repository link, and presentation materials.

All files should follow the prescribed folder structure and naming conventions to ensure smooth evaluation.

Participants are encouraged to publish their completed project on GitHub and showcase it on LinkedIn as part of their professional portfolio.

PROJECT SUCCESS CRITERIA

“The dashboard should not only display charts and KPIs.

It should answer business questions.

It should generate meaningful insights.

It should support business decision-making.

It should demonstrate professional dashboard design and storytelling.”